

The Impact of AI Investment News on Big Tech Stock Prices

DSA 210 Term Project · Ayberk Arpacı · Revised September 2026

This is a revised version of the report submitted on 30.05.2025. The analysis notebooks were corrected after submission (a volatility test that compared volatility with zero, target leakage in the machine learning models, forward-filled market holidays and pipeline code that did not match the data). Every number below comes from the current notebooks in the repository. The original report is kept in the git history.

1. Motivation

Companies announce AI investments constantly, and the financial press covers them daily. Whether this coverage moves stock prices is less clear. This project asks, for Tesla, Amazon, Google, Microsoft and Nvidia between January and April 2025:

- Does the tone of AI-related news predict next-day stock returns?
- Do days with positive news see better returns than days with negative news?
- Does volatility rise after AI news?
- Do the five companies react differently?
- Can technical indicators plus news sentiment predict tomorrow's price direction?

2. Data and Methodology

2.1 Data

Dataset	Contents	Source
Stock prices	Daily open, high, low, close, volume; 75 trading days (2 Jan to 22 Apr 2025)	Yahoo Finance (yfinance)
Indicators	Simple daily return, 10-day rolling volatility (standard deviation of returns), 14-day Wilder RSI, traded value (close × volume)	Computed
News	783 headlines for "<company> AI investment" (779 after removing duplicates)	Google News (GoogleNews library)
Sentiment	VADER compound score in [-1, 1] per headline; positive above 0.05, negative below -0.05	Computed

The only missing values are the warm-up rows of the rolling indicators, which are left empty. Four business days have no prices because US markets were closed (9 January, Martin Luther King Jr. Day, Presidents' Day and Good Friday); they are not filled in.

2.2 Event table and abnormal returns

- Each headline is mapped to the first trading session on or after its date. Headlines about the same company on the same day are averaged, giving **315 company-day events** (167 positive, 105 neutral, 43 negative).
- **Peer-adjusted (abnormal) return**: a stock's return minus the mean return of the five stocks that day. This removes sector-wide moves, most importantly the early-April tariff sell-off.
- For each event: next-day return, cumulative abnormal return over days 0 to +5 (CAR(0,+5)), and mean absolute return in the five days before and after.

A key property of the data: every company has AI news on 74 of its 75 trading days. There are no "quiet" days to compare against, so the effect of an announcement versus no announcement cannot be measured. The analysis compares the *tone* of the news instead.

2.3 Statistical tests

Spearman rank correlation, OLS regression with heteroskedasticity-robust (HC3) standard errors, Welch t-tests, a paired t-test and a Wilcoxon signed-rank test, one-way ANOVA and Kruskal-Wallis. Tests are one-sided where the hypothesis has a direction, and a Holm correction is applied across all of them (alpha = 0.05).

2.4 Machine learning

Target: whether tomorrow's close is above today's. Features, all known at today's close: return, volatility, RSI, change in log volume, mean return of the five stocks, mean sentiment and number of articles for the day, and the ticker. After dropping warm-up rows there are 305 rows over 61 trading days. The first 80% of days train the models (23 Jan to 1 Apr) and the last 20% test them (2 to 21 Apr); a random split would leak future information into training. Models: logistic regression, random forest, XGBoost, SVM (RBF) and KNN tuned with time-series cross-validation, compared with a majority-class baseline and in a four-fold walk-forward evaluation.

3. Findings

3.1 Descriptive overview

Company	Total return	Mean daily return	Mean volatility	Mean RSI	Mean headline sentiment
Tesla	-37.3%	-0.49%	0.050	39.2	0.106
Nvidia	-28.5%	-0.34%	0.045	44.0	0.067
Amazon	-21.4%	-0.29%	0.024	43.4	0.208
Google	-19.2%	-0.26%	0.023	41.2	0.108
Microsoft	-12.2%	-0.15%	0.019	43.0	0.127

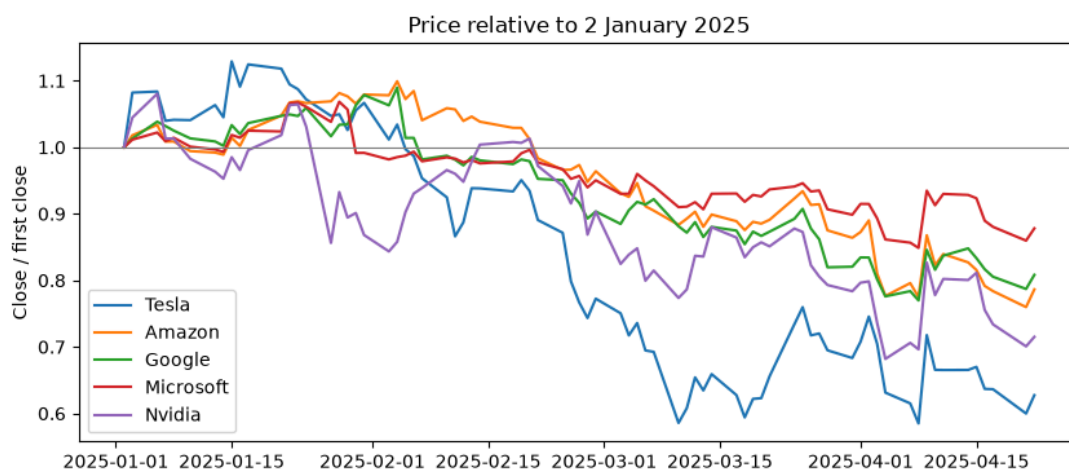


Figure 1. Price relative to 2 January 2025. All five stocks fell, most sharply in early April.

Most headlines are neutral (score exactly 0) or mildly positive; negative headlines are rare.

3.2 Hypothesis tests

	Hypothesis (alternative)	Result	Holm p	Decision
H1	Higher sentiment, higher next-day abnormal return	Spearman $\rho = +0.148$	0.022	Reject H0
H2	Positive-news days beat negative-news days next day	+0.27% vs -0.92%, Welch $t = 2.84$	0.019	Reject H0
H2b	Positive-news days beat neutral days next day	+0.27% vs -0.05%, $t = 1.15$	0.142	Cannot reject
H3	Volatility is higher in the 5 days after news than before	Mean $ return $ 2.38% before, 2.53% after	0.088 to 0.142	Cannot reject
H4	CAR(0,+5) differs across companies	ANOVA $F = 3.42$; Kruskal-Wallis $H = 16.8$	0.015 to 0.037	Reject H0

An OLS regression of next-day abnormal return on sentiment gives a slope of 0.012 (robust $p = 0.009$): raising a day's average sentiment by 0.8, from neutral to strongly positive, goes with about one percentage point better relative performance the next day, but sentiment explains only a small share of daily variation.

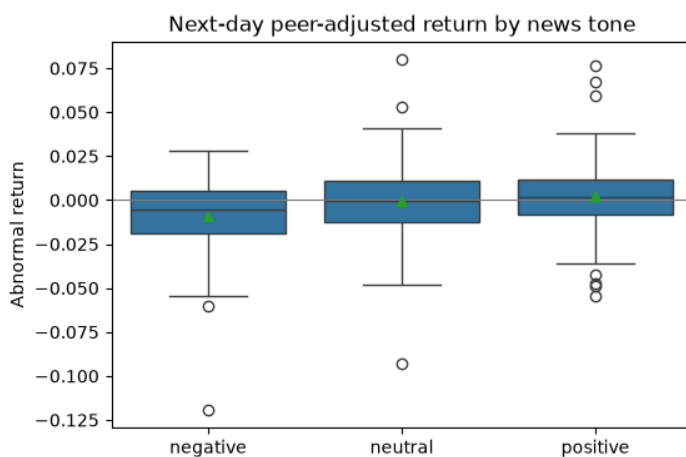


Figure 2. Next-day peer-adjusted return by the tone of the day's news.

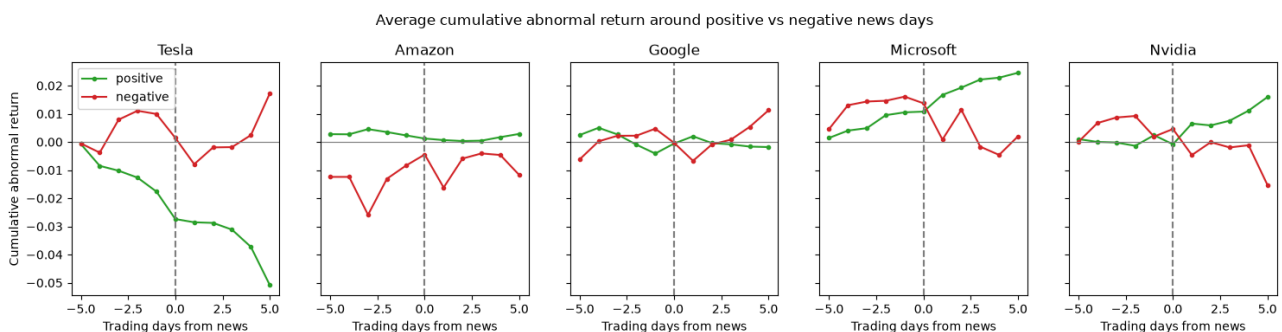


Figure 3. Average cumulative abnormal return from 5 days before to 5 days after positive and negative news days.

Average 5-day abnormal returns after news days were +1.1% for Microsoft, +0.3% for Google and Nvidia, +0.2% for Amazon and -1.9% for Tesla.

3.3 Machine learning

Model	Accuracy	Balanced accuracy	ROC AUC (test)	ROC AUC (walk-forward mean)
Majority baseline	0.677	0.500	0.500	-
Logistic Regression	0.462	0.528	0.590	0.636
Random Forest	0.615	0.691	0.699	0.650
XGBoost	0.585	0.656	0.648	0.604
SVM (RBF)	0.554	0.621	0.658	0.668
KNN (tuned)	0.646	0.627	0.656	0.595

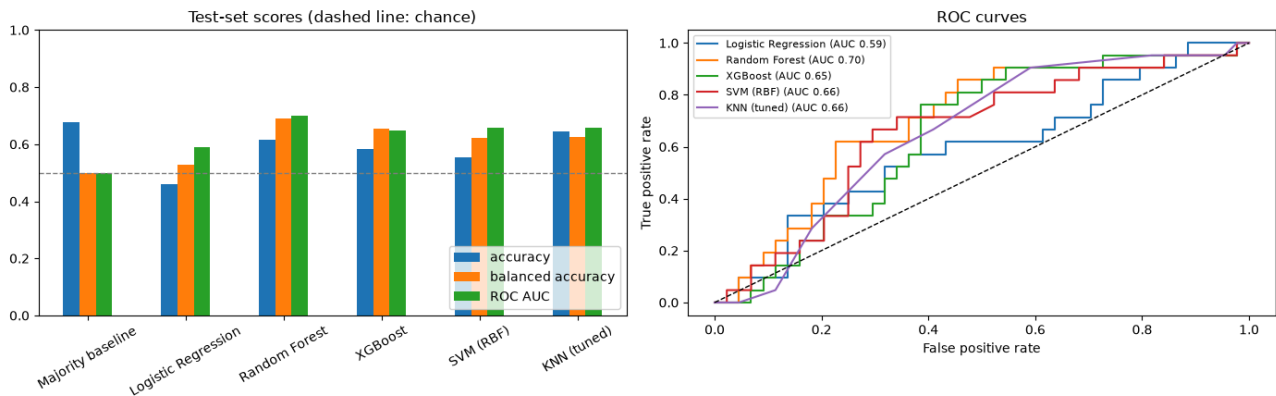


Figure 4. Test-set scores and ROC curves.

- No model beats the majority baseline on accuracy. The test period was dominated by the tariff sell-off, so predicting "down" was right two times out of three.
- Ranking quality is modest: test ROC AUC of 0.59 to 0.70, and walk-forward means of 0.60 to 0.67 with wide variation between folds (0.45 to 0.79).
- The random forest relies mainly on the sector return, volume change and the stock's own return, which suggests short-term mean reversion. Sentiment and article count rank lower.
- The first version of this project reported up to 87% accuracy. It predicted the same day's close-above-open while using that day's open and close as inputs, with a random split. The notebook reproduces that setup (80.7% accuracy) to show that the score came from leakage.

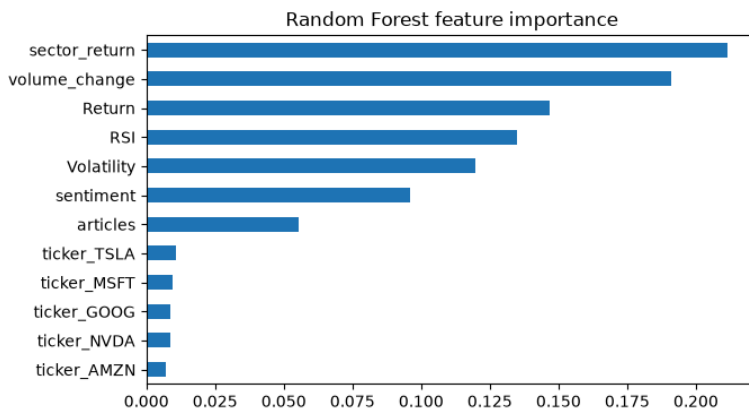


Figure 5. Random forest feature importance.

4. Limitations

- News appears on almost every trading day, so announcement versus no announcement cannot be compared.

- Only headlines are scored, with a general-purpose lexicon (VADER) that misses finance-specific language.
- Publication times are unknown; an article is treated as known at the close of its event day.
- Event windows of consecutive days overlap, so observations are not fully independent and the p-values are somewhat optimistic.
- Four months of data from a falling market; 61 usable trading days for the models, with the five stocks moving together.

5. Future Work

- Several years of data and a broader set of companies, including sectors outside tech.
- Article timestamps, so that news can be matched to intraday or next-session returns precisely.
- A finance-specific sentiment model (for example FinBERT) applied to full article text.
- A market index benchmark (for example the S&P 500 or Nasdaq-100) alongside the peer adjustment.
- Distinguishing news types (earnings, product launches, capital spending) instead of treating all AI news the same.

6. Conclusion

Within this sample, the tone of AI news carries a small, statistically significant signal: days with positive coverage are followed by slightly better returns than days with negative coverage, relative to the other large tech stocks. There is no evidence that AI news raises volatility, and next-day direction cannot be predicted better than a naive baseline. The companies did differ, with Microsoft holding up best and Tesla worst. Given the short, turbulent period and the near-daily news flow, these results describe early 2025 rather than a general law, and they do not support a trading strategy.

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